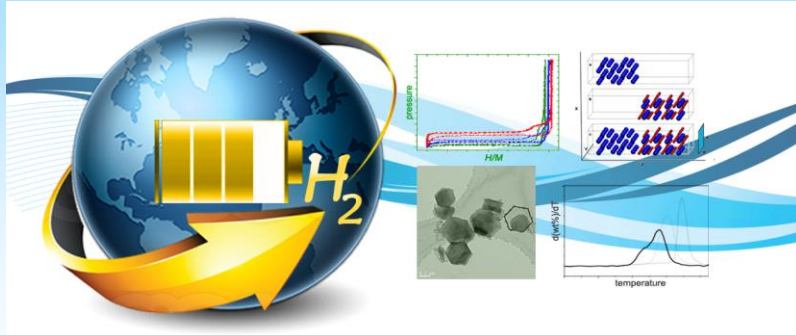


# Nanostructured materials for solid state hydrogen storage



## COST Action MP1103

Action Chair: Amelia Montone (ENEA, Italy)

Research aim:

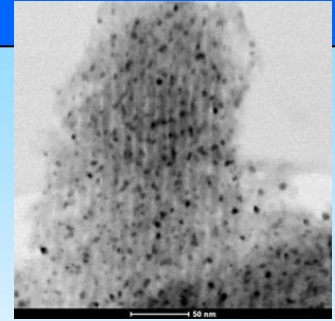
Develop innovative **nanostructured materials** that meet the targets for practical **Solid State Hydrogen Storage** (SSHS) for their adequate implementation in stationary and mobile **energy storage** applications

Participants: 26 Countries, ~ 250 members

# Organisation: four working groups

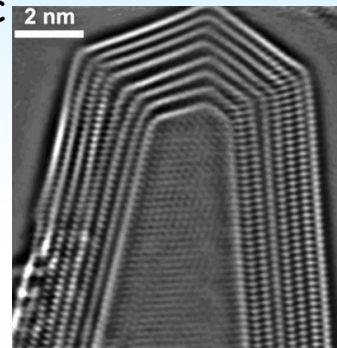
- WG1: **Synthesis** of novel materials with optimized properties

Group Leader: **Fermín Cuevas** (CNRS, France)



- WG2: High resolution and high sensitivity **characterization** of atomic level structure and of microstructural features

Group Leader: Sara Blas (Belgium)



- WG3: Characterization of **hydrogen storage properties** both at the laboratory level and at the scale of prototype tanks

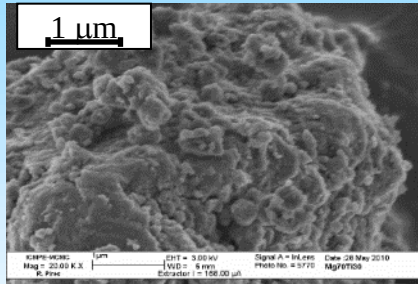
Group Leader: Martin Dornheim (HZG, Germany)



- WG4: **Computational modeling** of processes relevant to SSHS

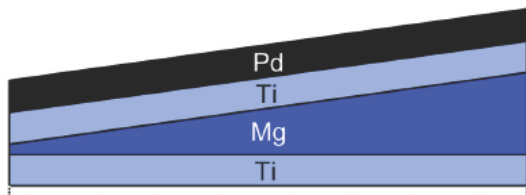
Group Leader: Tejs Vegge (Denmark)

## Management of Synthesis WG: architecture - based



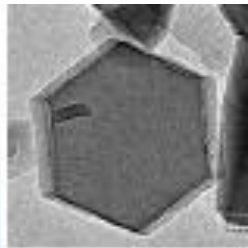
## Task 1: Nanostructured bulk materials.

Leader: Torben Jensen  
(Aarhus University, Denmark)



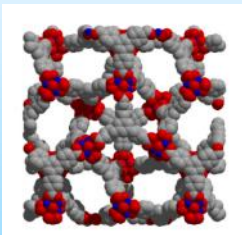
## Task 2: Thin films.

Leader: Bernard Dam  
(Delft University of Technology, Netherlands)



### Task 3: Nanoparticles and core-shell structures.

Leader: Kondo-Francois Aguey-Zinsou  
(University of New South Wales, Australia)



## Task 4: Porous and nanoscaffold hybrid materials.

Leader: Michael Hirscher  
(Max Planck Institute for Intelligent Systems, Germany)



# Outline

- ❖ Introduction: Mg for solid-state hydrogen storage
- ❖ Synthesis: Reactive ball milling under hydrogen
- ❖ Applications:
  - $\text{MgH}_2\text{-TiH}_2$  for solid-state hydrogen storage
  - $\text{Mg}_2\text{TMH}_x$  (TM = Fe, Co, Ni) for Li-ion batteries
- ❖ Conclusions

# Universal material requirements for energy storage

- Reversibility under operating conditions
- High specific and volumetric energy densities
- Fast loading and unloading kinetics
- Long cycle life
- Environmentally friendly
- Safety
- Low-cost



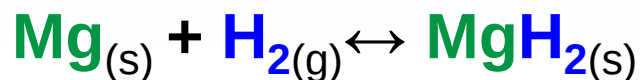
# Magnesium for solid-state hydrogen storage



## Advantages

Abundant, low-cost, light-weight

Simple hydrogen reaction



High storage capacity : 7.6 wt.% H, 109 gH<sub>2</sub>/L



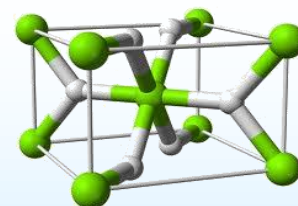
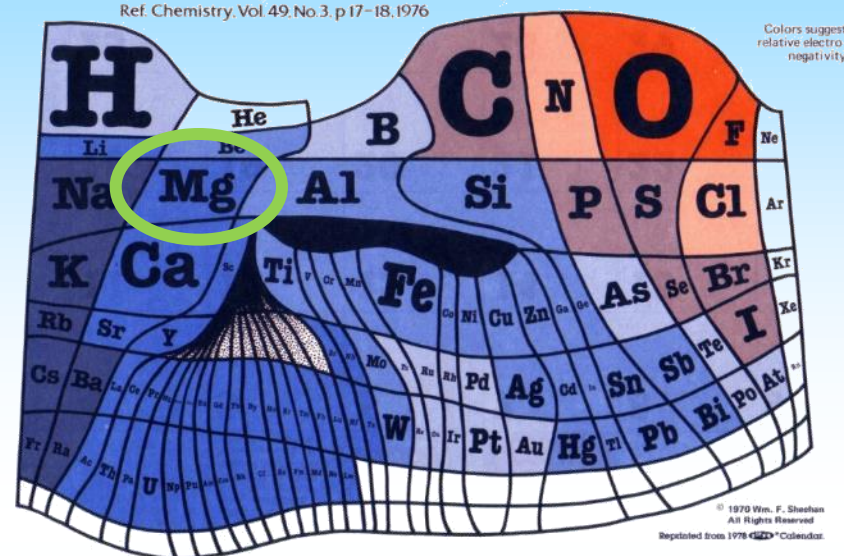
## Drawbacks

Too stable hydride for RT applications :  $P_{\text{eq}, \text{H}_2} = 1 \text{ atm at } 280^\circ \text{C}$

**Slow sorption kinetics**

## The Elements According to Relative Abundance

A Periodic Chart by Prof. Wm. F. Sheehan, University of Santa Clara, CA 95053  
Ref. Chemistry, Vol. 49, No. 3, p. 17-18, 1976

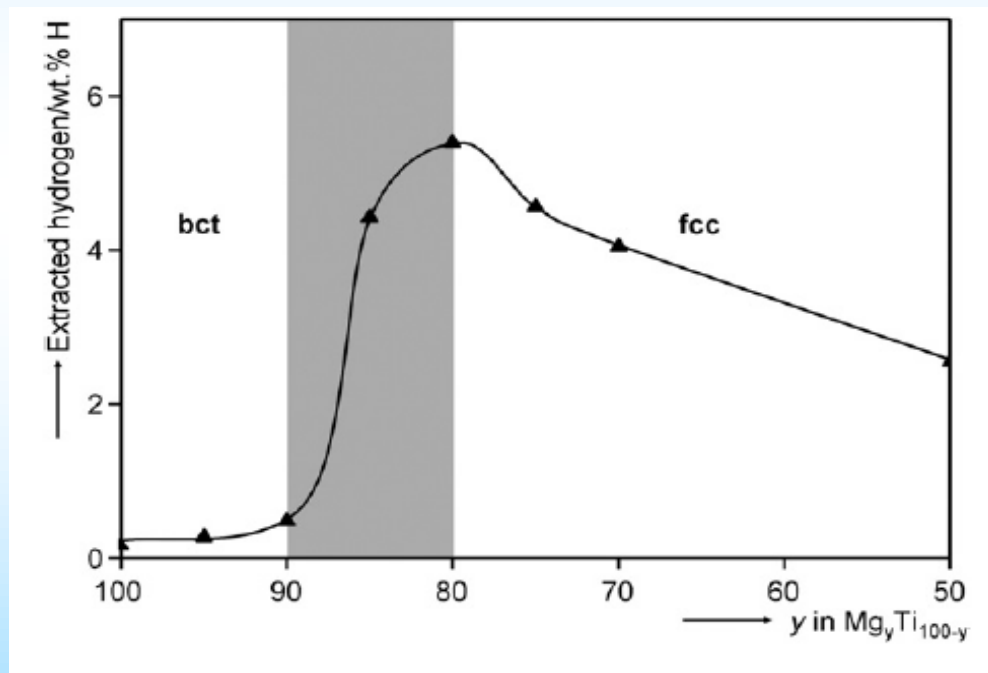


# Applications: Ni-MH electrochemical storage

- Negative electrode of Ni-MH battery  $\text{Mg} + x\text{H}_2\text{O} + xe^- \rightleftharpoons \text{MgH}_x + x\text{OH}^-$

✓ Kinetics enhancement: Ti-incorporation in Mg lattice

transition from bct-rutile ( $\text{MgH}_2$ ) to fcc-fluorite ( $\text{TiH}_2$ ) structure

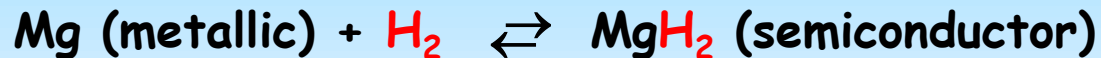


*Vermeulen et al., J. Mater. Chem., 2008, 18, 3680*

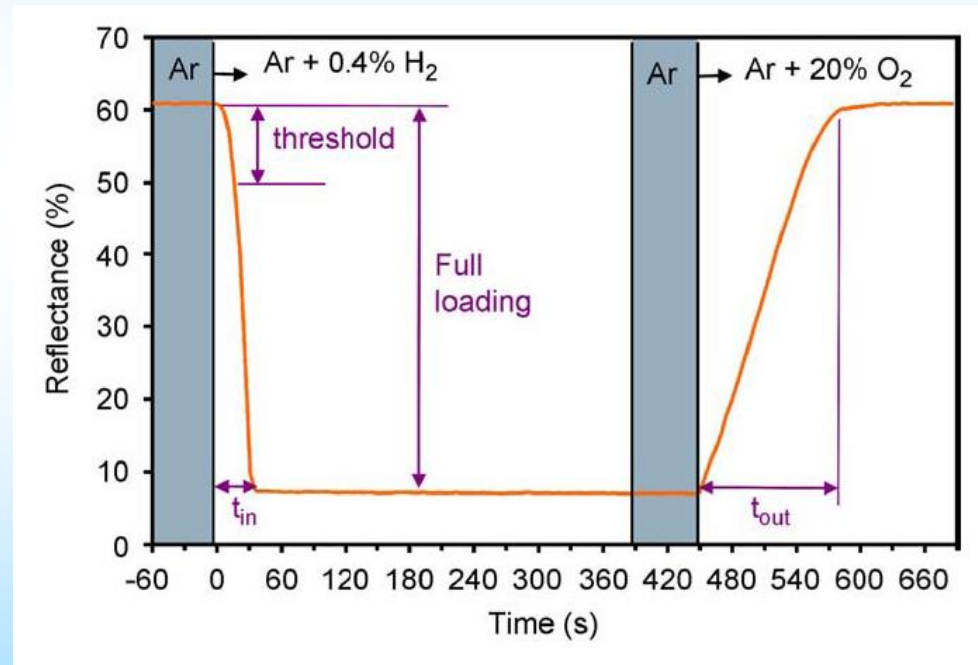
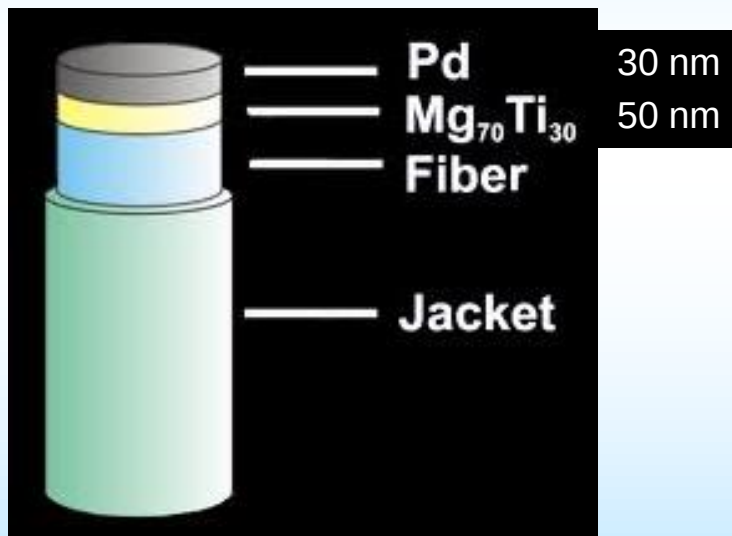


# Applications: hydrogen sensors

- Solid-gaz hydrogen absorption induces a metal to semiconductor transition



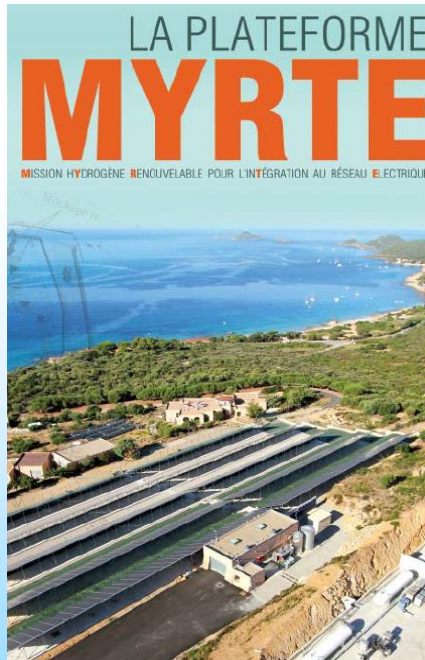
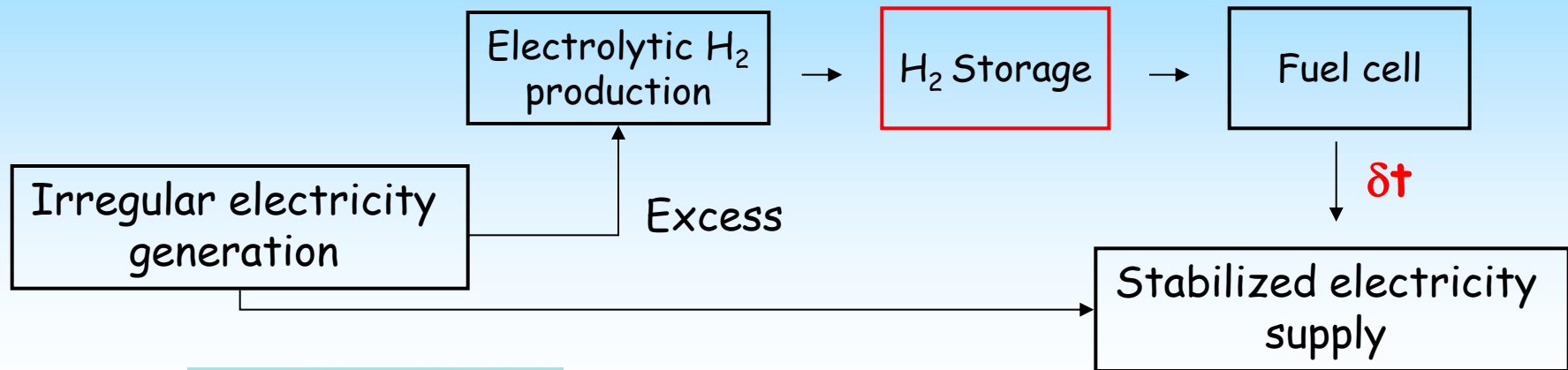
- ✓ Kinetics enhancement: thin film technology, Ti-incorporation, Pd coverage



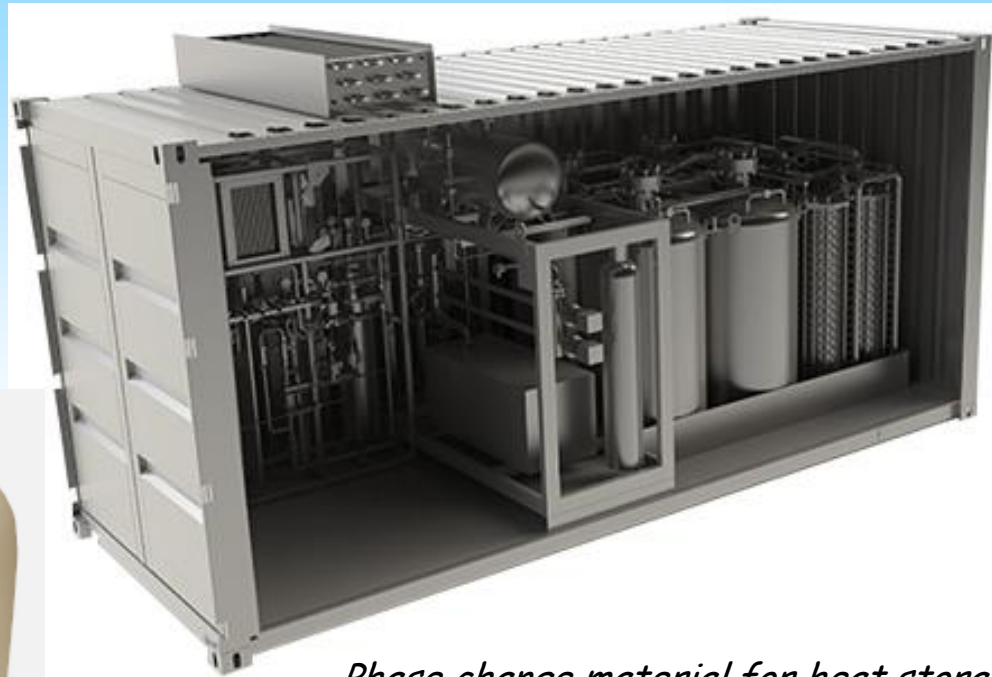
*Slaman et al., Sens. Actuators B, 2007, 123, 538*

# Applications: hydrogen storage for grid regulation

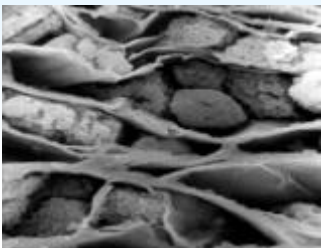
- Principle: regulation of electricity generation from intermittent sources



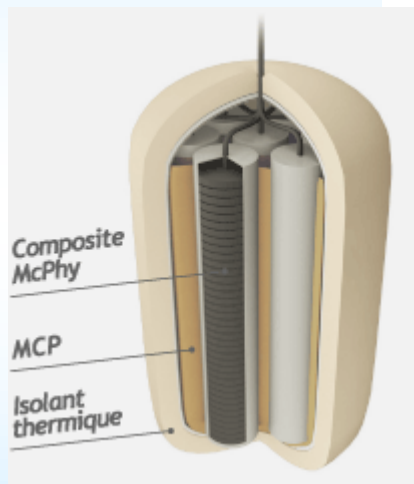
# Hydrogen storage: Mg-based nanohydride



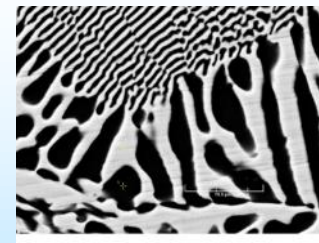
*Composite:  $MgH_2$  + graphite*



*Compressed composite*



*Phase change material for heat storage  
 $Mg_{69}Zn_{28}Al_3$*



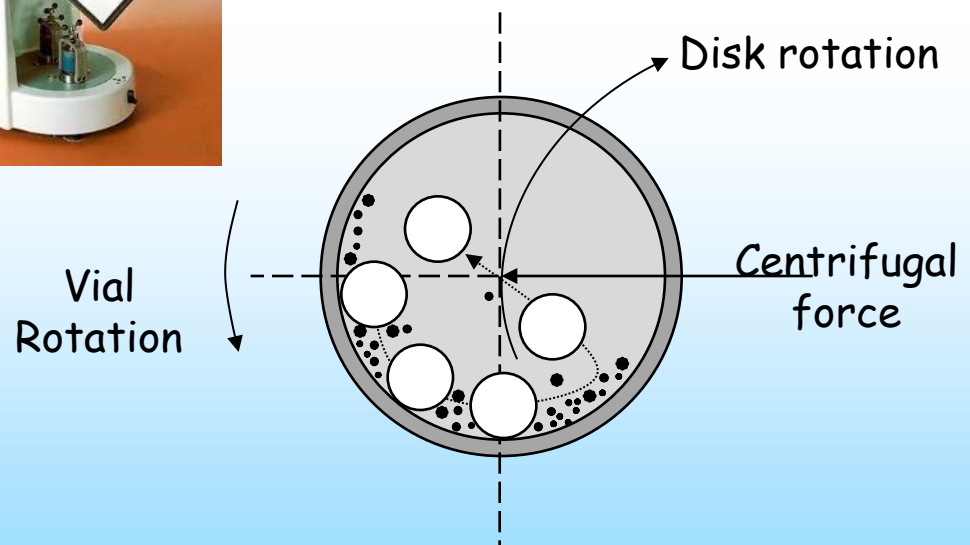
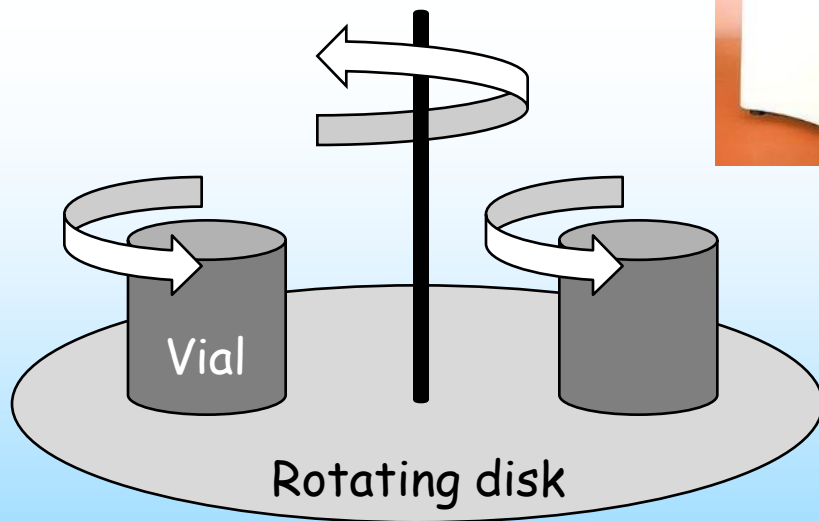
*Garrier et al., Int. J. Hydrogen Energy, 2013, 38, 9766*

✓ Kinetics enhancement: nanostructuration by milling, additives, temperature

## Reactive ball milling (RBM) under hydrogen

# Ball milling: the planetary principle

- Eccentrically vials rotating in opposite direction to the supporting disk.
- The planetary movement generates balls take-off.
- Balls collide between them and with vial walls, crushing by friction and impact the powders inside the vial.





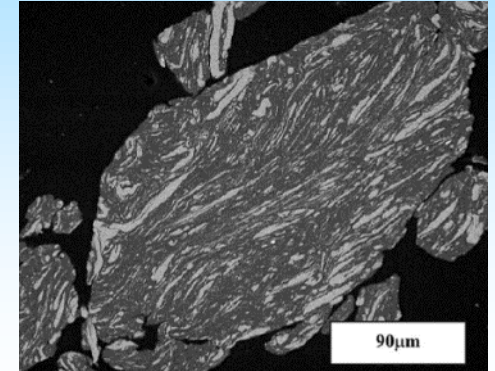
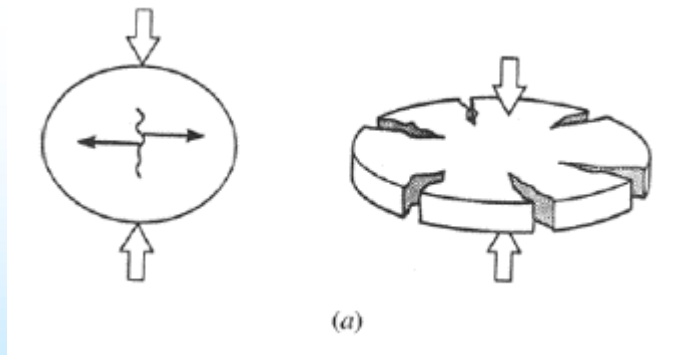
# Ball milling: nanostructuration

Repeated solid state **fragmentation** and **cold welding**

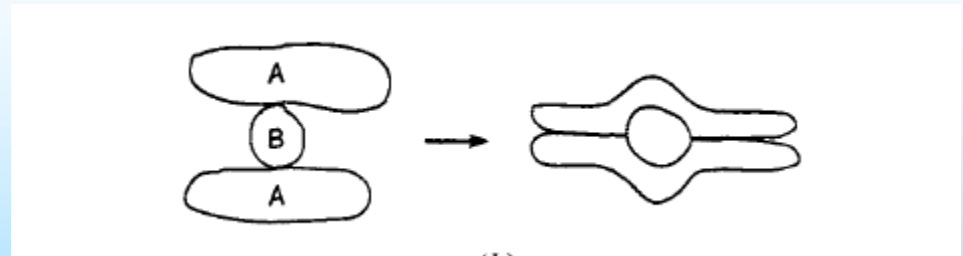
Synthesis of nanocomposites: Ni-superalloys / oxide

1970. J. Benjamin, *Met. Trans. 1* (1970) 2943

**Fracture**: fresh surfaces, size-reduction



**Welding**: solid-state reactions, mixing



D.R. Maurice and T.H. Courtney: *Metall. Trans. A*, 1990, vol. 25A, pp. 147-157

# RBM under hydrogen: fast synthesis of nanohydrides

→ Fracture on milling



Generation of fresh surfaces  
⇒ **Clean solid-gas interface**



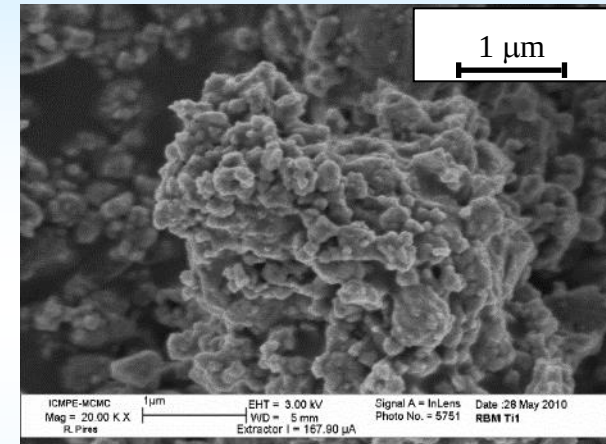
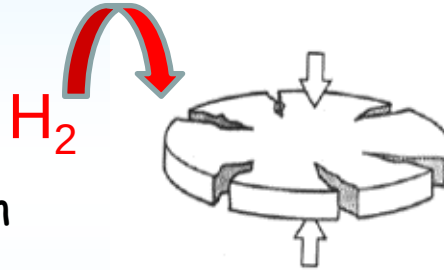
Hydrogen absorption



Hydrogen embrittlement

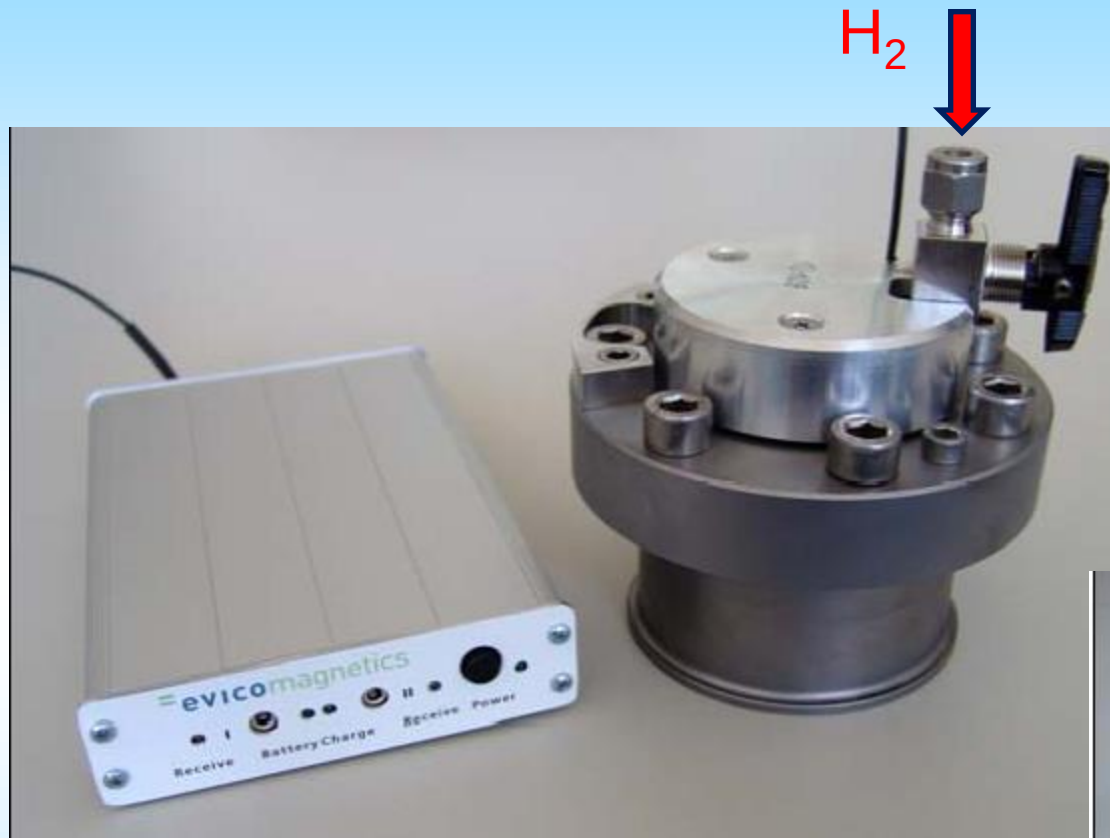


Particle size refinement



TiH<sub>2</sub> - nanohydride

# RBM under hydrogen : Evicomagnetics commercial device



$P_i(H_2) = 150 \text{ atm}$

Vial volume: 200 ml

Monitoring of P and T

$H_2$  in

P,T sensors

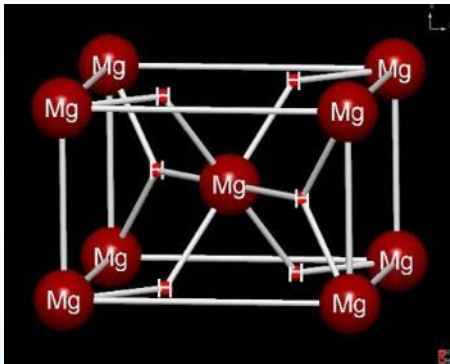
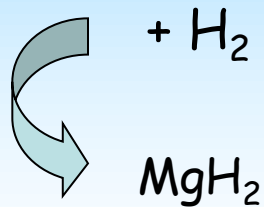


*S. Doppiu et al. J. Alloys Comp. 427 (2007) 204*

# $\text{MgH}_2\text{-TiH}_2$ nanocomposites for solid-state hydrogen storage

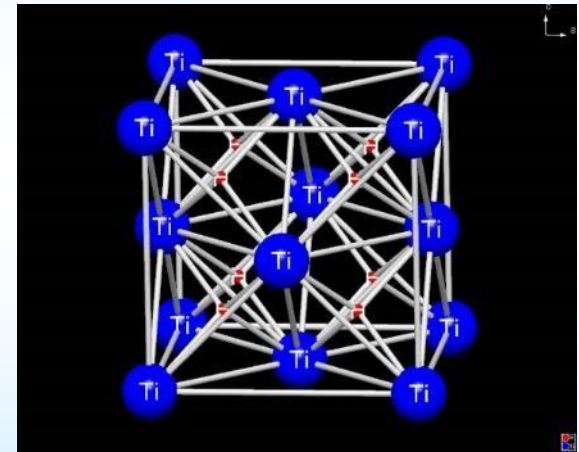
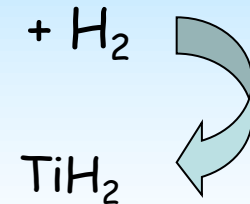
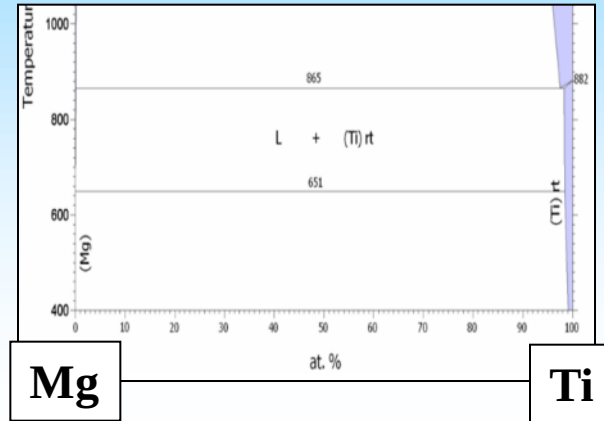
# Mg-Ti system and binary metal hydrides

- Mg-Ti: an immiscible system



$$D(300\text{ K}) \sim 10^{-12} \text{ cm}^2 \text{ s}^{-1}$$

Rutile-type structure



$$D(300\text{ K}) \sim 10^{-7} \text{ cm}^2 \text{ s}^{-1}$$

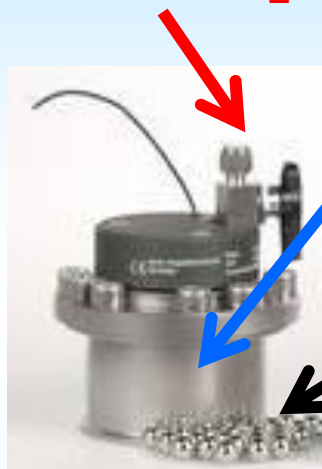
Fluorite-type structure



# Synthesis of $(1-y)\text{MgH}_2$ - $y\text{TiH}_2$ nanocomposites

- Evicomagnetics commercial vial: In-situ monitoring of P and T

8 MPa  $\text{H}_2$



Mg + Ti  
powders

Steel balls

Vial: hardened SS

Vial volume: 200 ml

Milling device: Fritsch P4

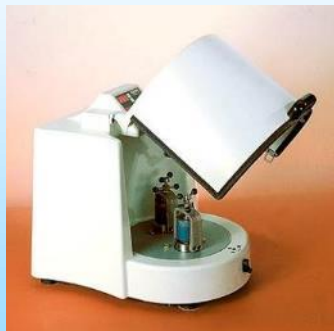
Disk speed: 400 rpm

Vial speed: - 800rpm vs. disk

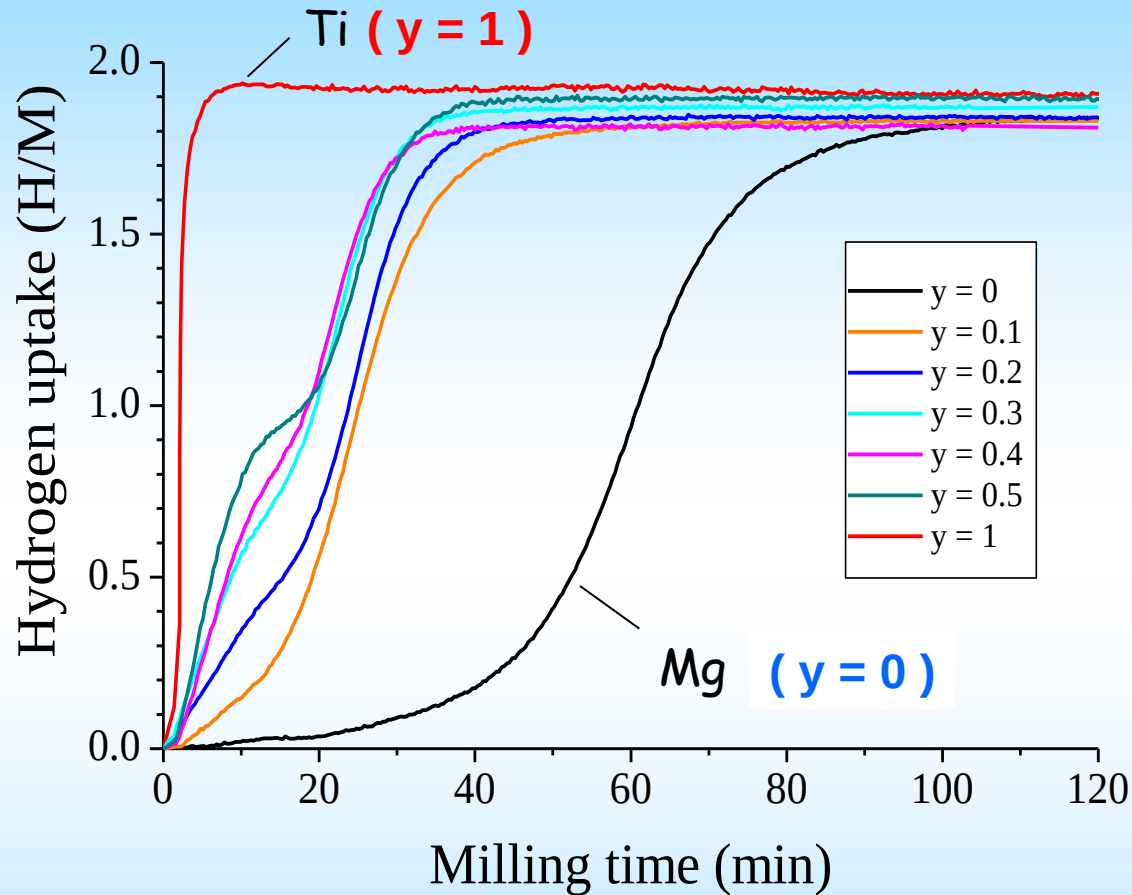
Balls: SS,  $\varnothing = 12$  mm

$m_{\text{metal powder}}: 5$  g

$m_{\text{Balls: powder}} = 60:1$



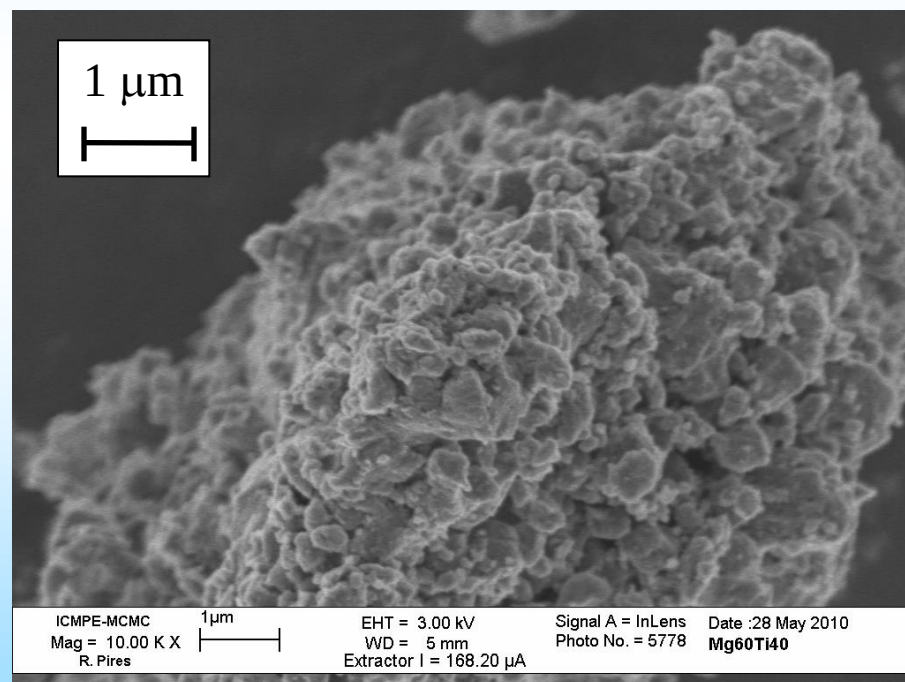
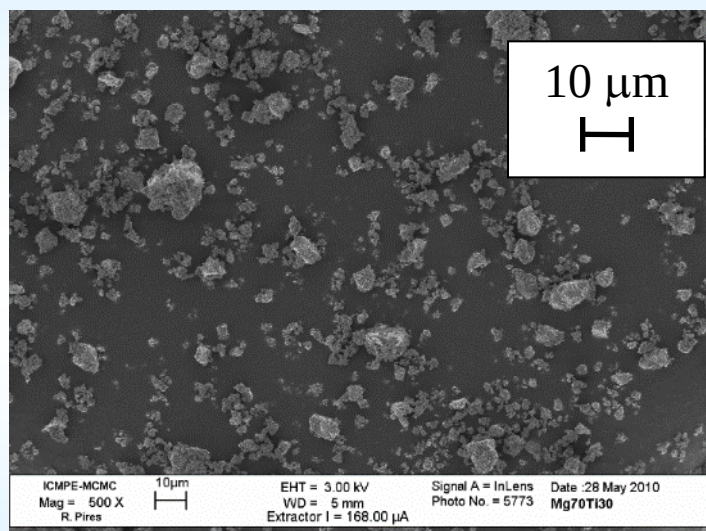
# In-situ absorption curves



- Fast nanocomposite synthesis ( $< 1\text{h}$ )
- Two steps: consecutive formation of  $\text{TiH}_2$  and  $\text{MgH}_2$  phases

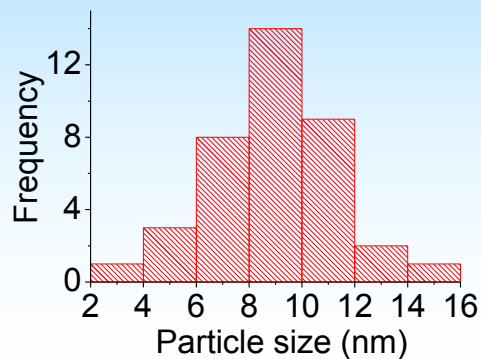
# MgH<sub>2</sub>-TiH<sub>2</sub> powder morphology: SEM analysis

- Reactants: Mg (< 800 μm), Ti (< 150 μm)
- Product: Micrometric-size agglomerates of nanoparticles



# MgH<sub>2</sub>-TiH<sub>2</sub> microstructure: TEM analysis

BF-mode



0.7MgH<sub>2</sub>-0.3TiH<sub>2</sub>

DF-mode  
(TiH<sub>2</sub> - selection)

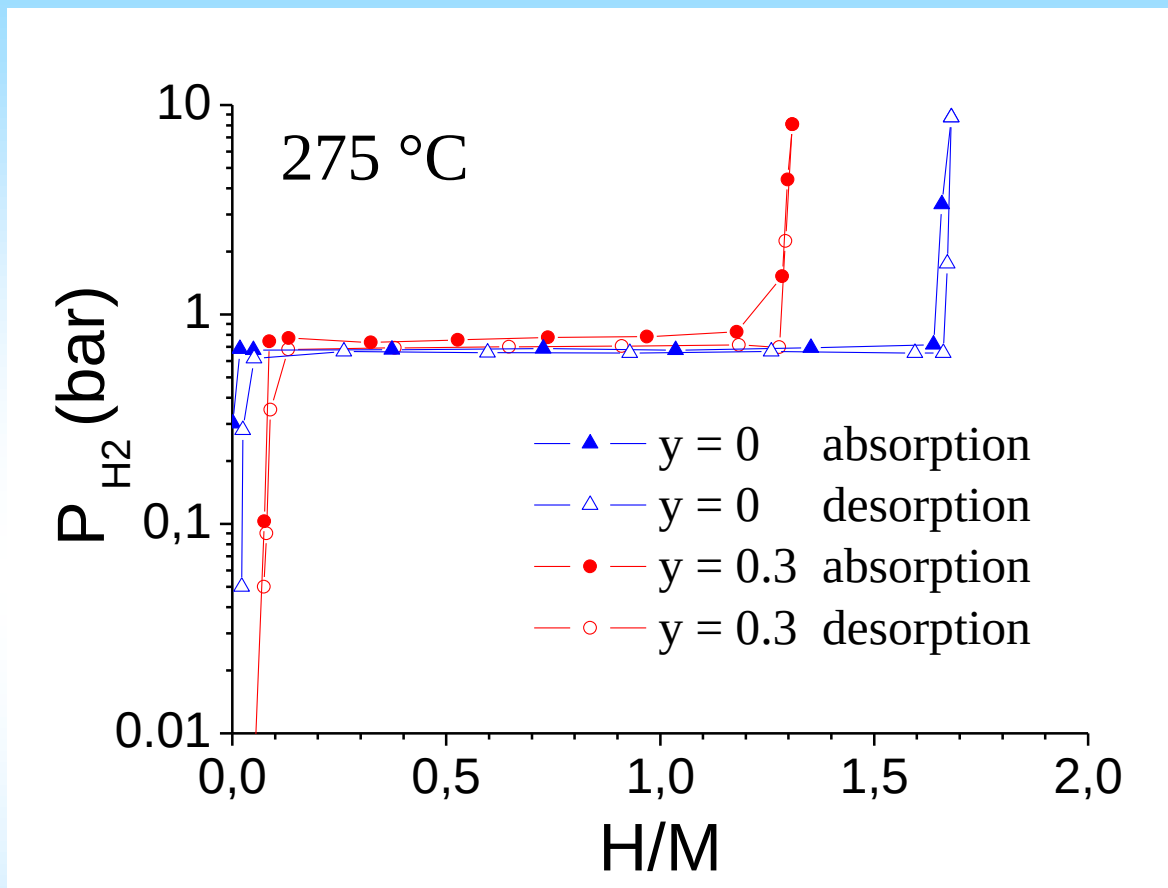


- Homogeneous mixing of MgH<sub>2</sub> and TiH<sub>2</sub> phases at nanoscale
- TiH<sub>2</sub> grain size: ~ 10 nm

Comparative study between

$\text{MgH}_2$  ( $y = 0$ ) and  $0.7\text{MgH}_2-0.3\text{TiH}_2$  ( $y = 0.3$ )

# H-thermodynamics

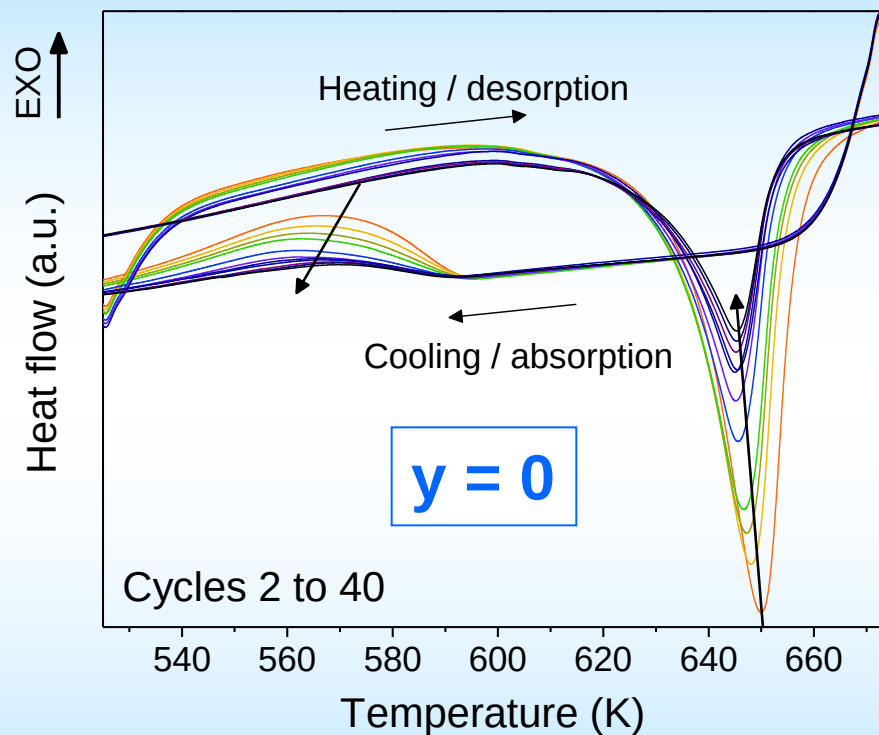


- Only hydrogen from  $MgH_2$  phase is reversibly stored
- No significant changes in Mg-H thermodynamics

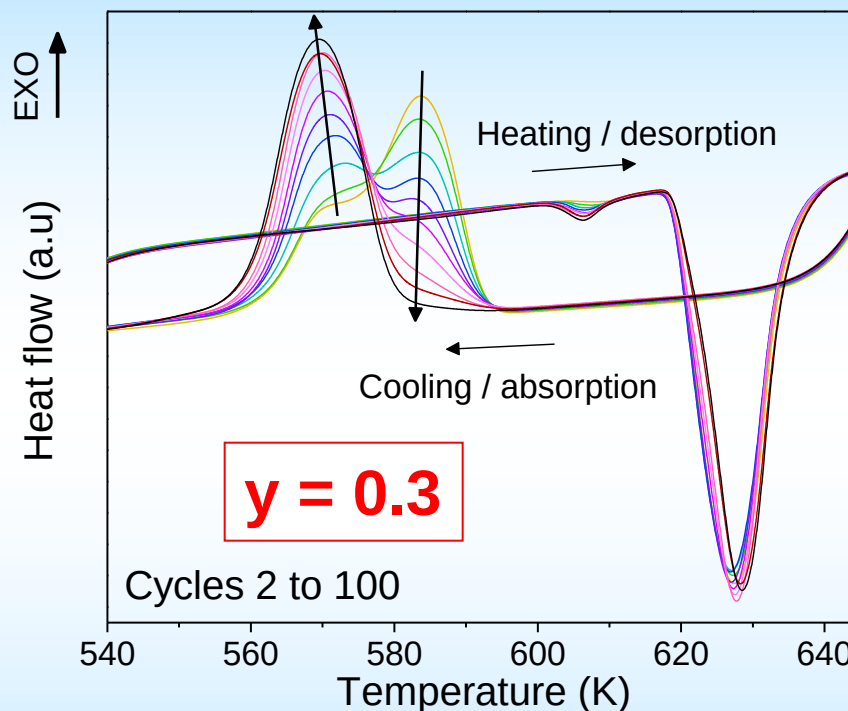


# Hydrogen cycling at constant pressure (0.4 MPa) : HPDSC

**MgH<sub>2</sub>**

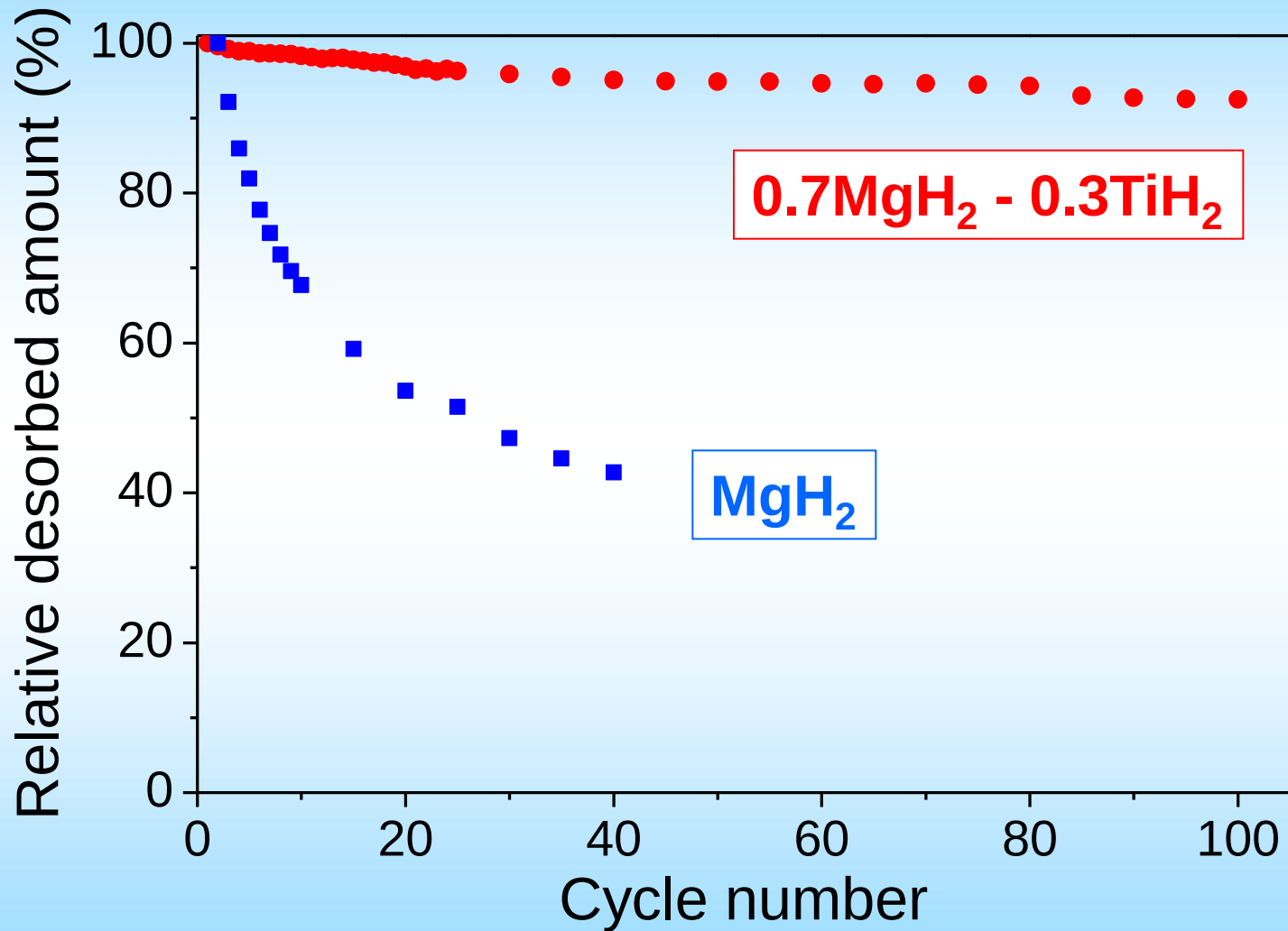


**0.7MgH<sub>2</sub> - 0.3TiH<sub>2</sub>**

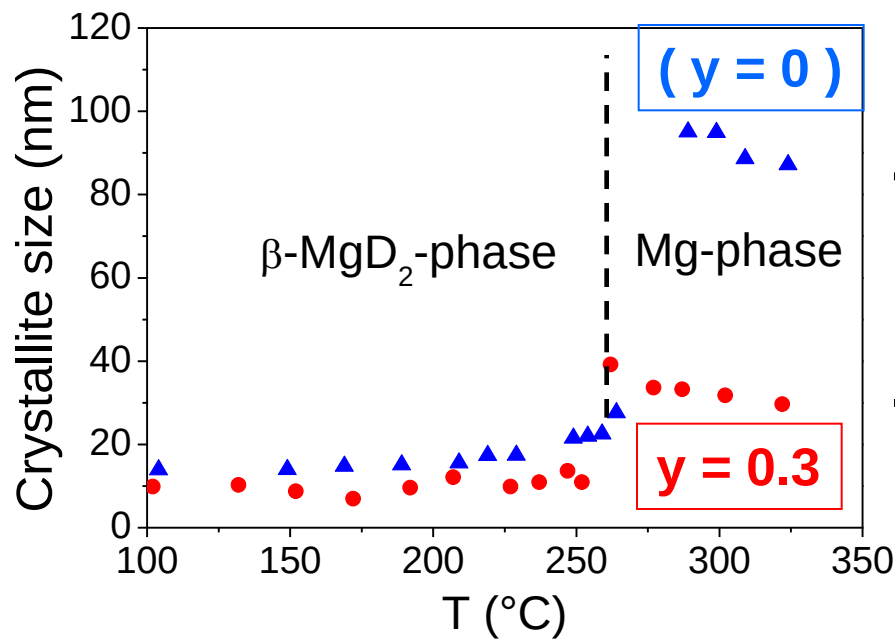
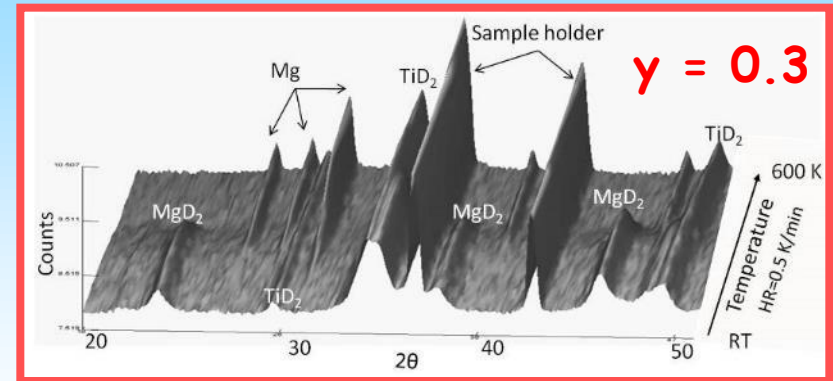
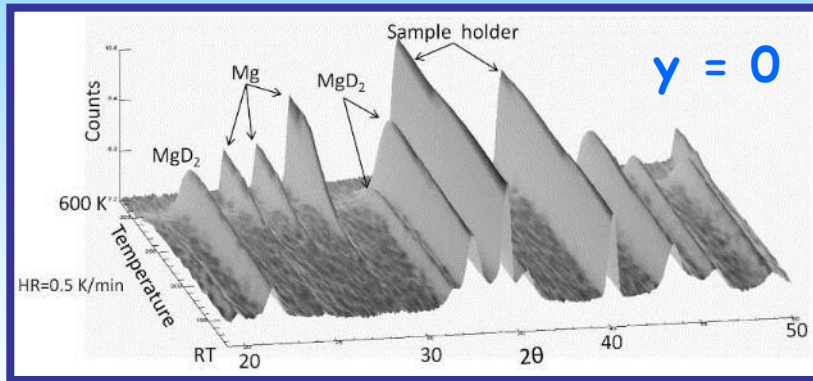


- Good reversibility for Ti-containing sample

# Hydrogen cycling at constant pressure (0.4 MPa) : HPDSC

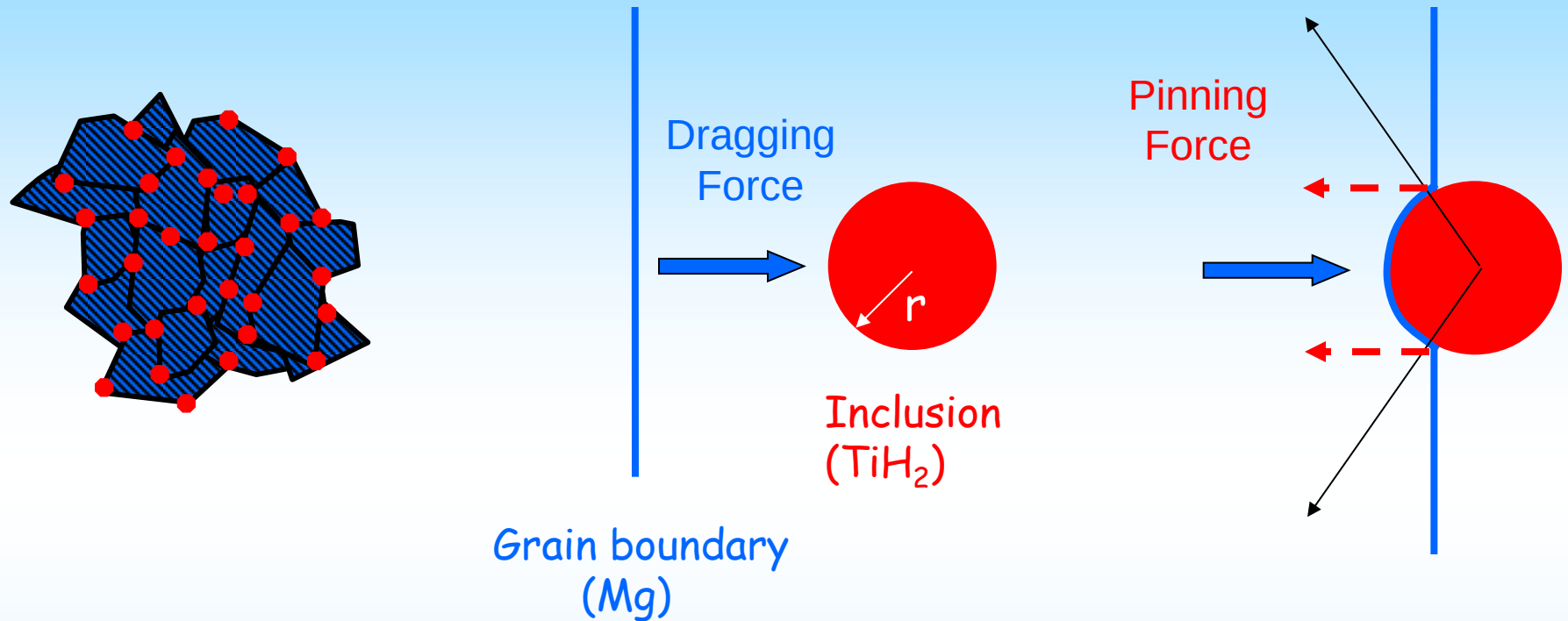


# Hindering of $\text{MgH}_2$ grain coarsening by $\text{TiH}_2$ inclusions



•  $\text{TiH}_2$  phase limits grain coarsening of Mg phase

# Grain coarsening: Zener pinning effect



- Limited grain growth:

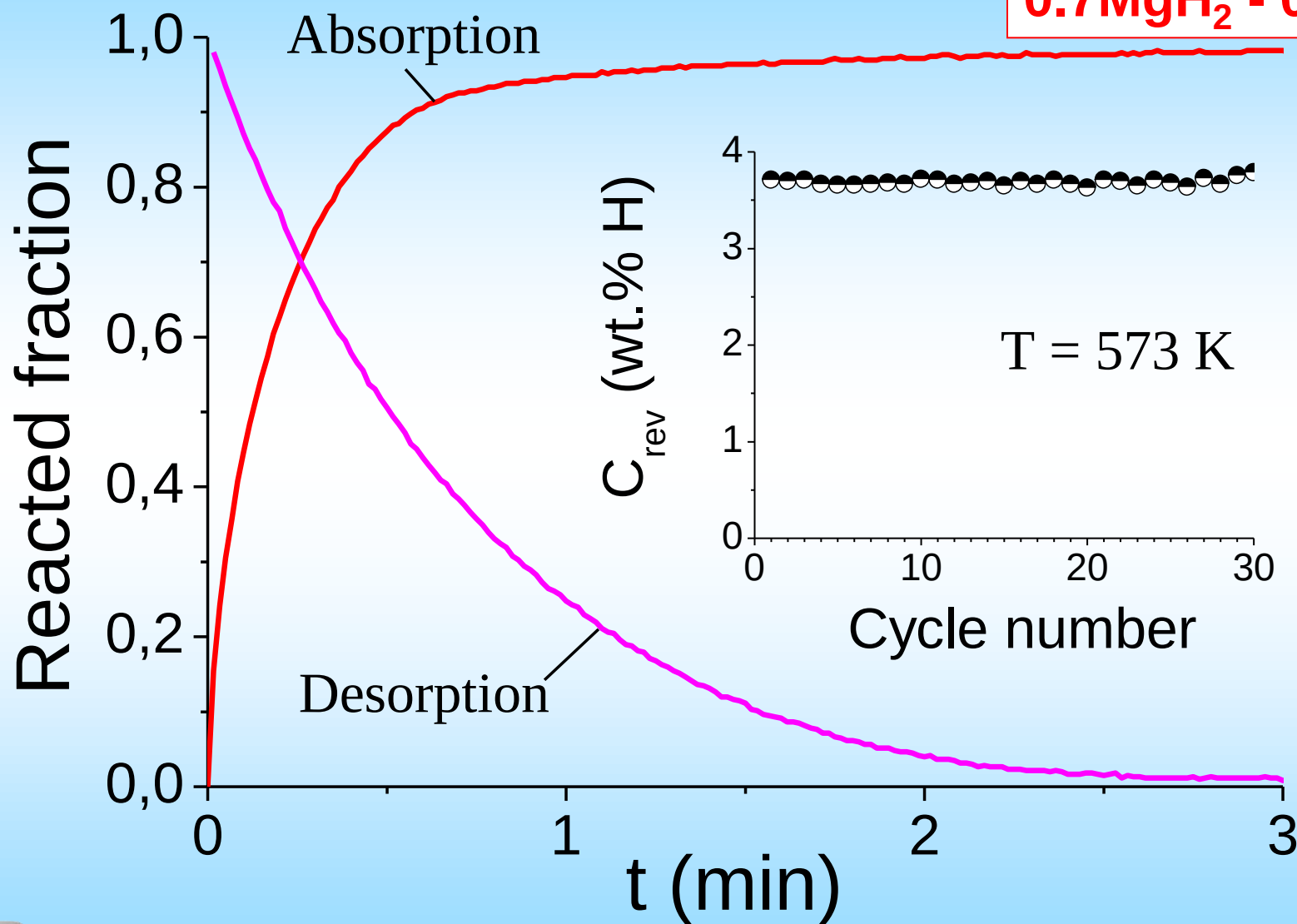
$$\langle L_c \rangle_{\max} = k \frac{r}{f}$$

$r$  = radius of inclusions

$f$  = volume fraction of inclusions

# MgH<sub>2</sub>-TiH<sub>2</sub> nanocomposites: fast and cycling-stable kinetics

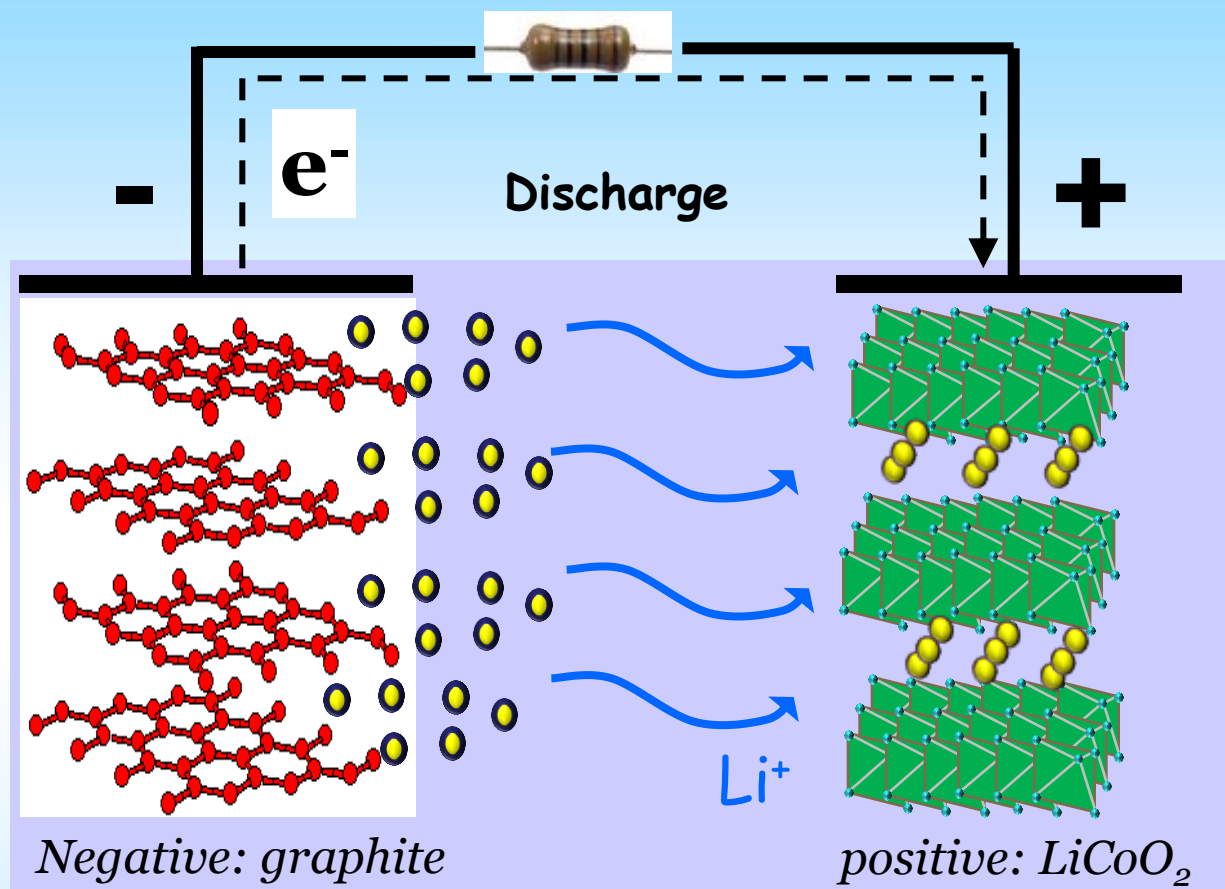
0.7MgH<sub>2</sub> - 0.3TiH<sub>2</sub>



$Mg_2TMH_x$  (TM = Fe, Co, Ni) nanohydrides for Li-ion batteries



# Classical Li-ion batteries: insertion reactions



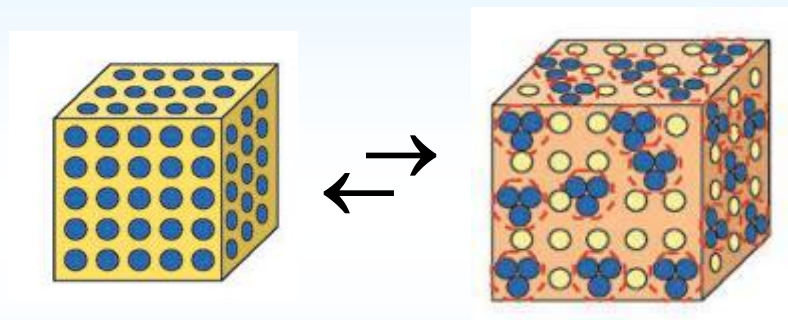
Negative electrode: graphite ( $\text{Li}_x\text{C}_6 \rightarrow 6\text{C} + x\text{Li}^+ + xe^-$ )

Positive electrode: lamellar oxides ( $\text{Li}_{1-x}\text{MO}_2 + x\text{Li}^+ + xe^- \rightarrow \text{LiMO}_2$ )

# Graphite substitution by conversion electrodes



$A = O, S, N, P, F...$

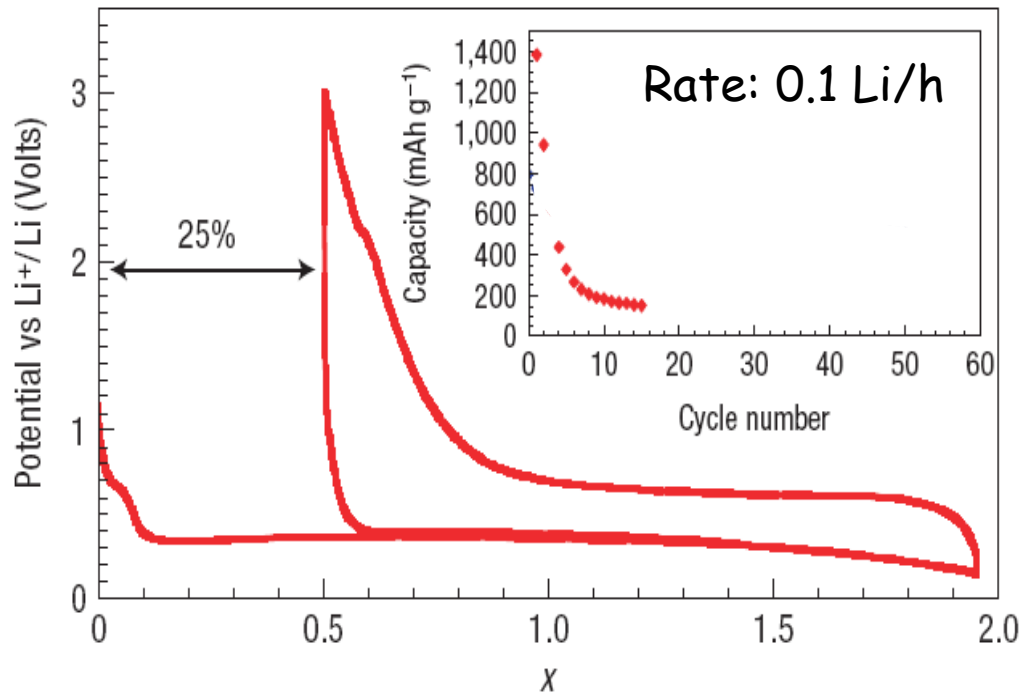


... and hydrides!



Oumellal et al. Nature Materials 7 (2008) 916

# Ball-milled $\text{MgH}_2$



Capacity: 2038  $\text{mAh/g}$

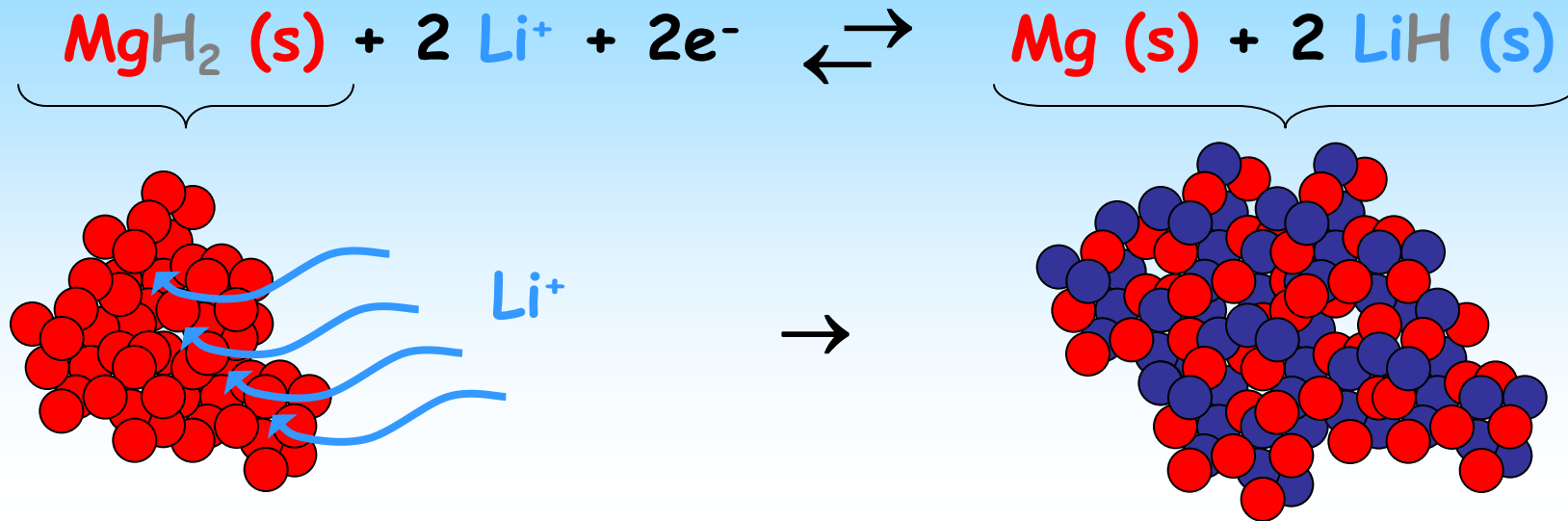
Reversibility: 75 %

Slow kinetics

Poor cycle life

Oumellal et al. Nature Materials 7 (2008) 916

# Challenges



Reversibility concerns (slow kinetics and poor cycle-life) due to:

Long-range mass transport at RT of Mg, Li and H species

Reversible breaking and forming of Mg-H and Li-H bonds

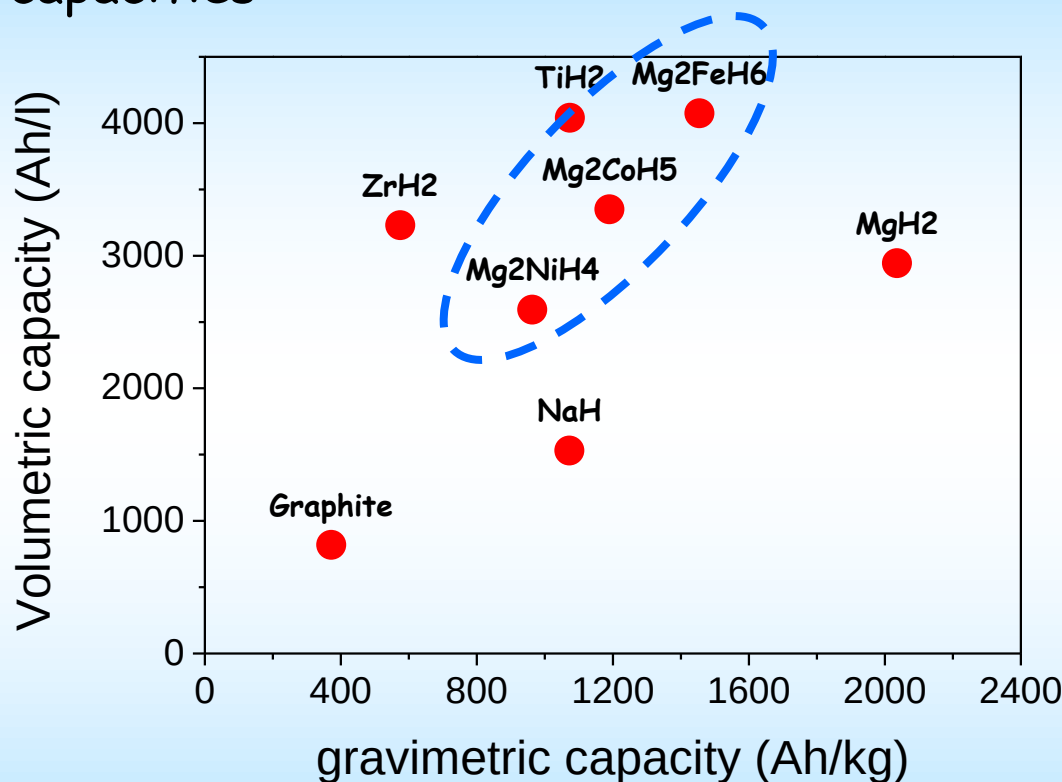
Strong volume changes within the electrode ( $\Delta V/V = 83\%$ )

Low electrical conductivity of MgH<sub>2</sub> and LiH phases

# Looking for better systems: $Mg_2TMH_x$ compounds

Why are they good candidates?

- Good theoretical capacities



- Transition metals (**TM**) may create an intrinsic **electronic percolation network**

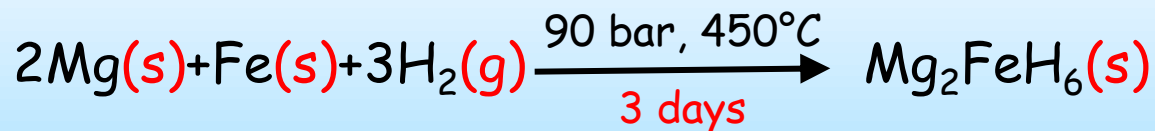
# Mg<sub>2</sub>TM (TM = Fe, Co, Ni) hydrides: synthesis & properties

- Complex hydrides: Covalent TM - H bonding (molecular complex)  
Ionic Mg<sup>2+</sup> - [TMH<sub>x</sub>]<sup>4-</sup> bonding

| Compound                         | H/M  | Cm<br>(wt.% H) | Cv<br>(g <sub>H</sub> /l) | T <sub>dec</sub><br>(°C) |
|----------------------------------|------|----------------|---------------------------|--------------------------|
| MgH <sub>2</sub>                 | 2    | 7.66           | 108                       | 280                      |
| Mg <sub>2</sub> FeH <sub>6</sub> | 2    | 5.47           | 150                       | 290                      |
| Mg <sub>2</sub> CoH <sub>5</sub> | 1.67 | 4.48           | 126                       | 320                      |
| Mg <sub>2</sub> NiH <sub>4</sub> | 1.34 | 3.62           | 98                        | 255                      |

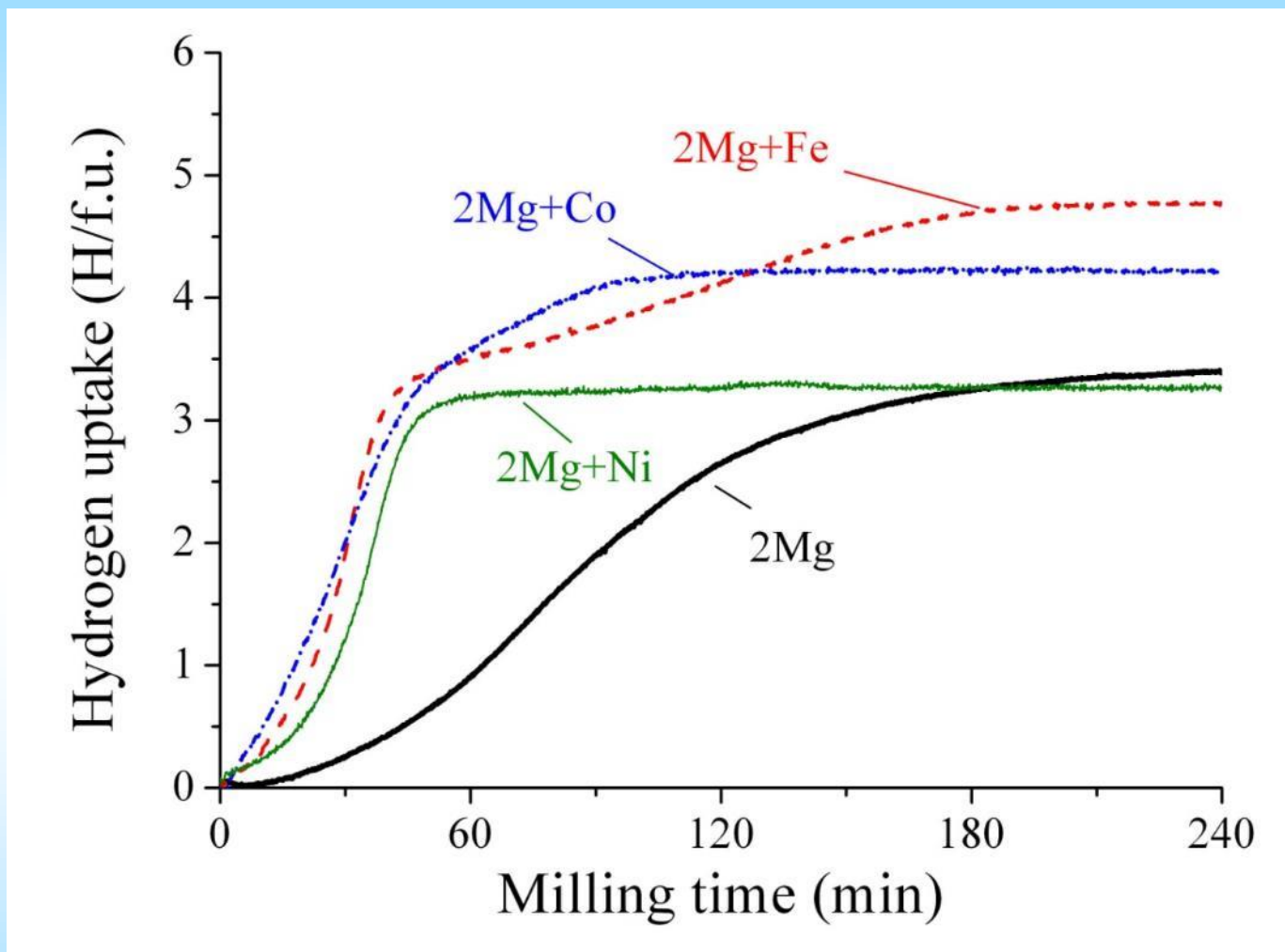


- Classical synthesis: sintering at high pressure and temperature



*Selvam and Yvon, IJHE 16 (1991) 615*

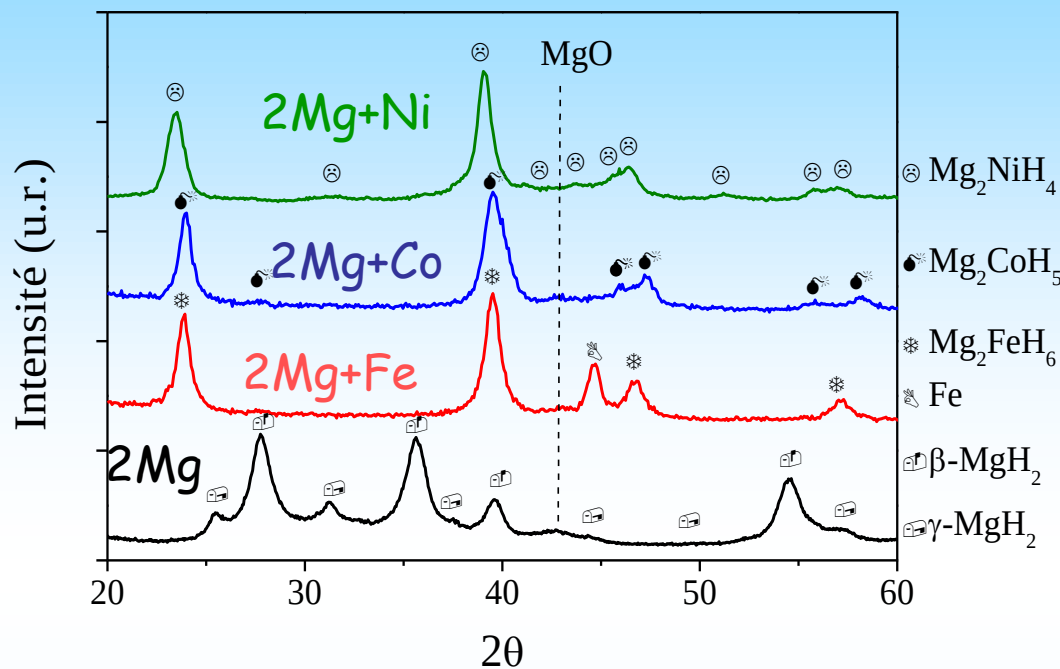
# Synthesis by reactive ball milling



Fast compound formation as compared to the classical sintering route



# Crystallographic studies of end products



| Reactants | Phases                    | S.G.       | Crystallite size |
|-----------|---------------------------|------------|------------------|
| Mg        | $\beta\text{-MgH}_2$      | $P4_2/mnm$ | $5 \pm 1$ nm     |
|           | $\gamma\text{-MgH}_2$     | $Pbcn$     | $3 \pm 1$ nm     |
| 2Mg+Fe    | $\text{Mg}_2\text{FeH}_6$ | $Fm-3m$    | $8 \pm 1$ nm     |
| 2Mg+Co    | $\text{Mg}_2\text{CoH}_5$ | $P4/mnm$   | $8 \pm 1$ nm     |
| 2Mg+Ni    | $\text{Mg}_2\text{NiH}_4$ | $C2/c$     | $9 \pm 1$ nm     |

# Electrochemical discharge (galvanostatic mode)

Swagelok half-cells

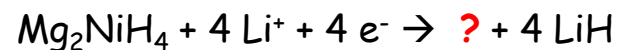
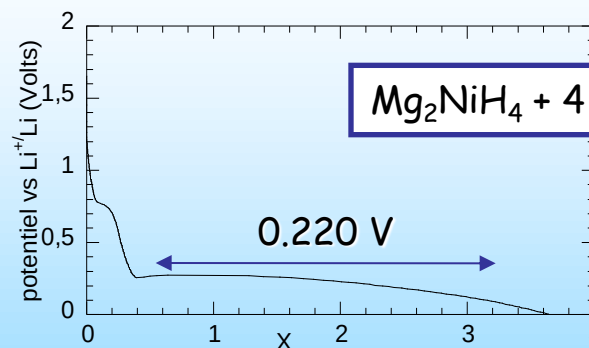
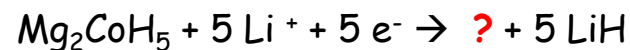
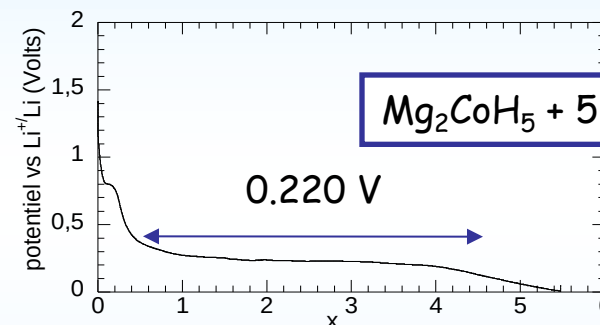
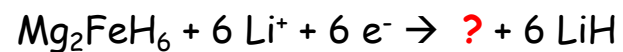
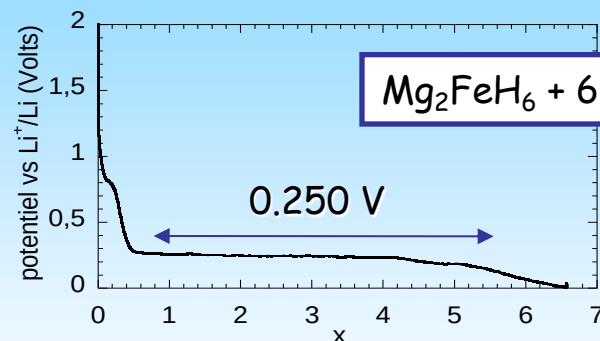
- Li



+  $\text{Mg}_2\text{TMH}_x$

Electrolyte: LP-30  
EC-DMC / 1M  $\text{LiPF}_6$

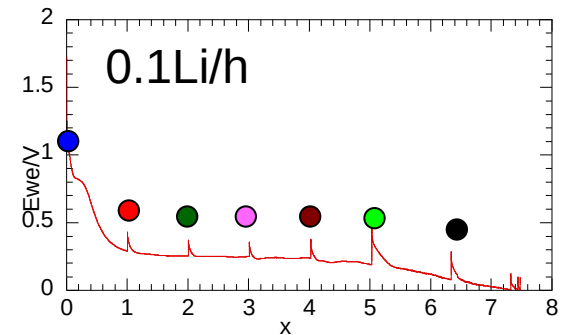
Rate: 0.1 Li/h



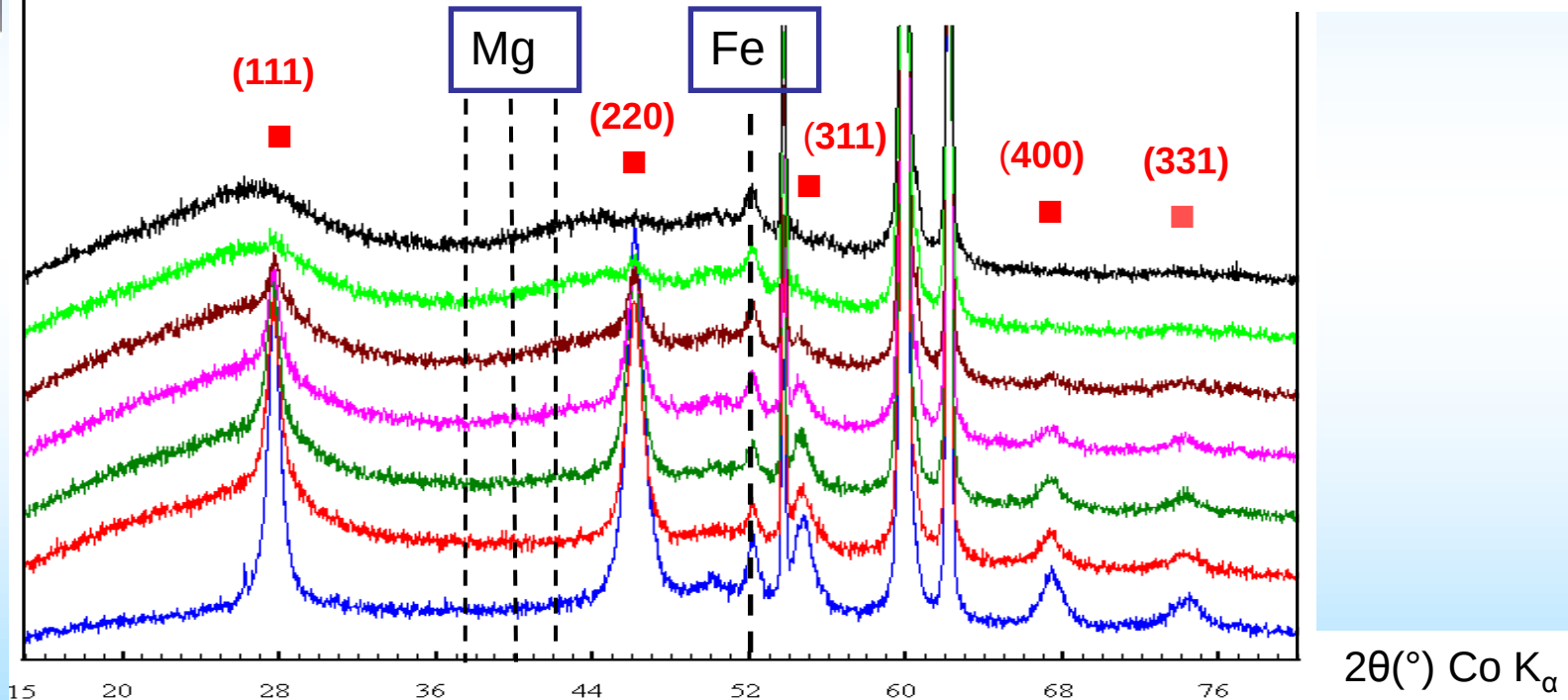
# Mg<sub>2</sub>FeH<sub>6</sub>: In-situ XRD studies (GITT on discharge)



■ Mg<sub>2</sub>FeH<sub>6</sub>

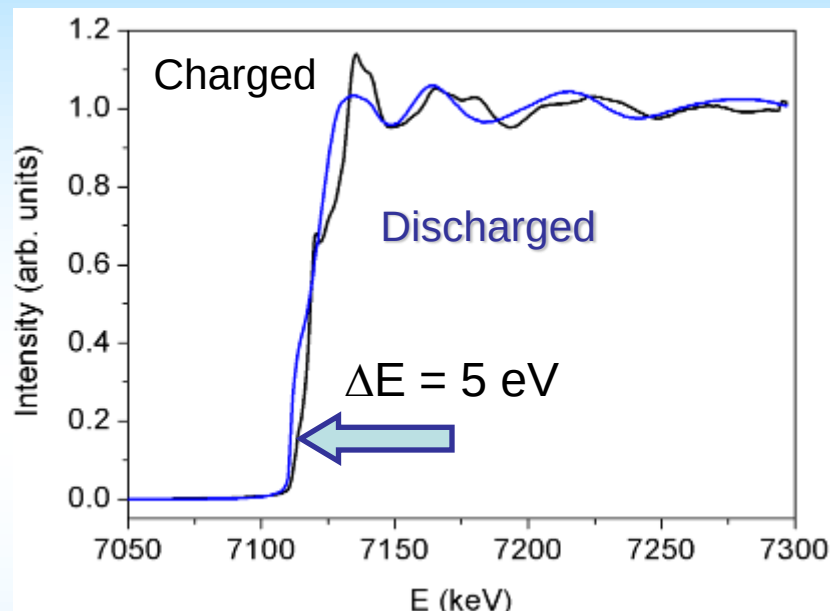


X=6.4 Li  
X=5 Li  
X=4 Li  
X=3 Li  
X=2 Li  
X=1 Li  
X=0 Li



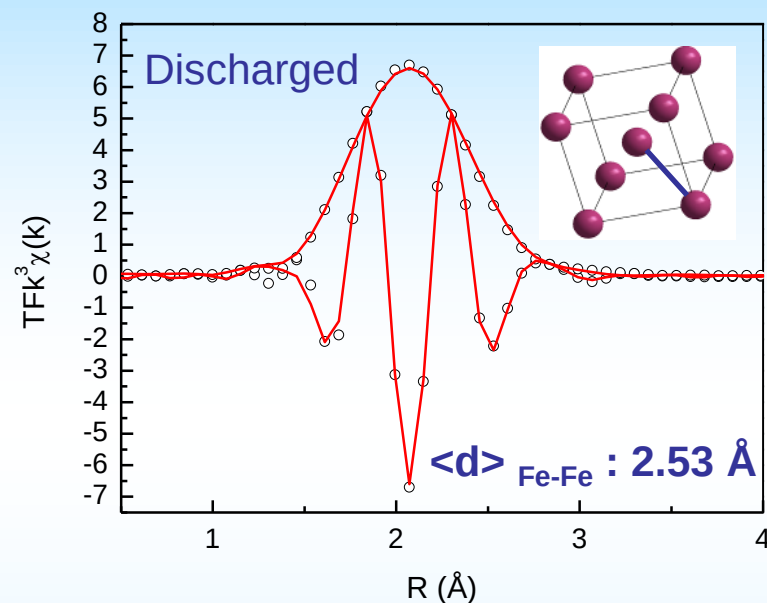
Two-phase transformation : Nano-Mg<sub>2</sub>FeH<sub>6</sub>  $\xrightarrow{+ 6 \text{ Li}^+ + 6 \text{ e}^-}$  Amph.-2Mg,Fe (+ 6 LiH)

## XANES spectra (Fe-K-edge)



Change of Fe-oxidation state:  $\text{Fe}^{2+} \rightarrow \text{Fe}^0$

## EXAFS refinement



Formation of amorphous Fe

- Discharged state: Amorphous Fe and Mg domains + LiH

# Conclusions

- ❖ Nanostructured Mg-based hydrides are efficient materials for energy storage
- ❖ They can be easily and rapidly synthesized in bulk form by reactive ball milling under  $H_2$  atmosphere
- ❖ They are excellent materials for reversible hydrogen storage at moderate temperatures and promising candidates for negative electrodes of Li-ion batteries

# Acknowledgements



- Junxian Zhang
- Marine Ponthieu
- Karine Provost
- Valérie Paul-Boncour



- José Francisco Fernández



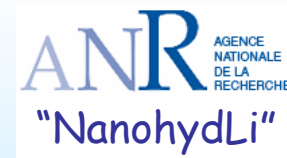
- Warda Zaïdi
- Jean-Pierre Bonnet
- Luc Aymard



- Laetitia Laversenne



- Stéphanie Belin



COST MP1103 workshop « Solid state hydrogen storage: **links between academia and industry** », Nantes, France. **12 May 2014**



IDHEA: International Discussion on Hydrogen Energy and Applications, Nantes, France. 12-14 May 2014





# Thank you for your attention!!!

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